Synthesis, Optical Characterization, and Photocatalytic Performance of CdS and Silica-Coated CdS Nanostructures

Abstract

Cadmium sulfide (CdS) powder was synthesized and systematically investigated to elucidate its structural, optical, and photocatalytic properties. X-ray diffraction analysis confirmed the formation of phase-pure crystalline CdS with a cubic zinc-blende structure, indicating good structural integrity of the synthesized material. Optical characterization using UV–visible spectroscopy combined with Tauc's relation revealed a direct optical band gap of 3.57 eV for CdS and 3.72 eV for CdS composite, which is significantly larger than that of bulk CdS and is attributed to nanoscale and surface-related effects. The photocatalytic performance of CdS was evaluated through the degradation of Rhodamine 6G under irradiation, and the degradation process followed a pseudo-first-order kinetic model. The apparent rate constants increased with the initial dye concentration, yielding values of 0.001, 0.0021, and 0.0075 min^{-1} for 10, 15, and 20 ppm, respectively. These results demonstrate that nanoscale CdS exhibits effective photocatalytic activity governed by its crystalline structure, modified electronic properties, and favorable surface reaction kinetics, highlighting its potential for light-driven environmental remediation applications.

Research Aim:

To synthesis and characterize CdS nanostructures and study its photocatalytic activity on Rhodamine 6G dye.

Research Objectives:

- Synthesis of CdS samples using simple synthesis route
- Apply a silica coating on the prepared samples
- Study the characterization of both bare CdS and Silica-coated CdS
- Study the photocatalytic activity of silica-coated CdS on Rhodamine 6G dye as a model of pollutant degradation

1. Introduction and Background

Semiconductor photocatalysis has attracted considerable attention as a sustainable approach for environmental remediation and solar energy conversion since the pioneering work on photoelectrochemical water splitting [1]. The fundamental principles and environmental applications of semiconductor photocatalysis have been extensively investigated, demonstrating its potential for pollutant degradation and energy-related processes [2,3]. The photocatalytic efficiency of semiconductor materials is strongly influenced by their electronic structure, light absorption capability, and surface reaction dynamics [4].

Cadmium sulfide (CdS), a II–VI group semiconductor with a direct band gap, has emerged as a promising material due to its strong optical absorption and efficient photogenerated charge carrier production [5–7]. Compared to wide-band-gap oxide photocatalysts, CdS can be activated under lower-energy irradiation, making it attractive for light-driven catalytic applications [8]. Consequently, CdS has been widely explored in photocatalysis, optoelectronics, and photoelectrochemical systems.

Various synthesis routes have been developed to prepare CdS with controlled structural and morphological characteristics, including chemical precipitation, hydrothermal processing, sol–gel methods, and vapor-phase techniques [9–11]. At the nanoscale, CdS exhibits modified optical and electronic properties due to size confinement and surface effects, which can significantly influence photocatalytic behavior [12,13].

Despite its favorable properties, CdS suffers from photocorrosion and structural instability under illumination, which limit its long-term performance [10,14]. To address these challenges, surface modification strategies have been widely investigated. Among them, silica (SiO_2) coating has proven effective due to its chemical inertness, optical transparency, and ability to suppress photocorrosion while preserving the intrinsic electronic structure of CdS [15]. Additionally, silica coating improves particle dispersion, reduces agglomeration, and enhances adsorption of reactant molecules, further contributing to improved functional performance.

As a result of these advantages, CdS and silica-modified CdS materials have been applied in diverse fields, including photocatalytic degradation of organic pollutants, solar energy conversion, and optoelectronic devices [5–8,15]. Continued development of synthesis and surface engineering approaches remains essential for optimizing the performance and stability of CdS-based materials in practical applications.

Photocatalysis

Photocatalysis is a process where light energy is used to accelerate a chemical reaction. This technique is particularly promising for addressing environmental and energy challenges, such as water purification, air cleaning, and hydrogen production. It is widely applied in environmental cleanup, and energy production from water splitting.

In photocatalysis, nanocomposites are used to improve the efficiency of photocatalysts. They offer several advantages:

- Enhanced charge separation: Nanocomposites help in better separation of photogenerated electron-hole pairs, which is crucial for improving photocatalytic efficiency1.
- Increased light absorption: They can absorb a broader spectrum of light, including visible light, which is more abundant than UV light.
- Improved stability: Nanocomposites often exhibit higher chemical and photochemical stability, making them more durable

The Photocatalysis mechanism is illustrated in Figure 1. The photocatalytic degradation mechanism using a CdS_Silica composite under visible light irradiation:

- **Light** excites electrons (e^-) in CdS_Silica across its band gap (**Eg = 3.67 eV**), generating electron-hole pairs (e^-/h^+).
- **Electrons** reduce oxygen (O_2) to form reactive oxygen species.
- **Holes** oxidize water or hydroxide ions to produce hydroxyl radicals (OH).
- These reactive species attack and break down organic pollutants into harmless products like **H_2O, CO_2**, and other byproducts.

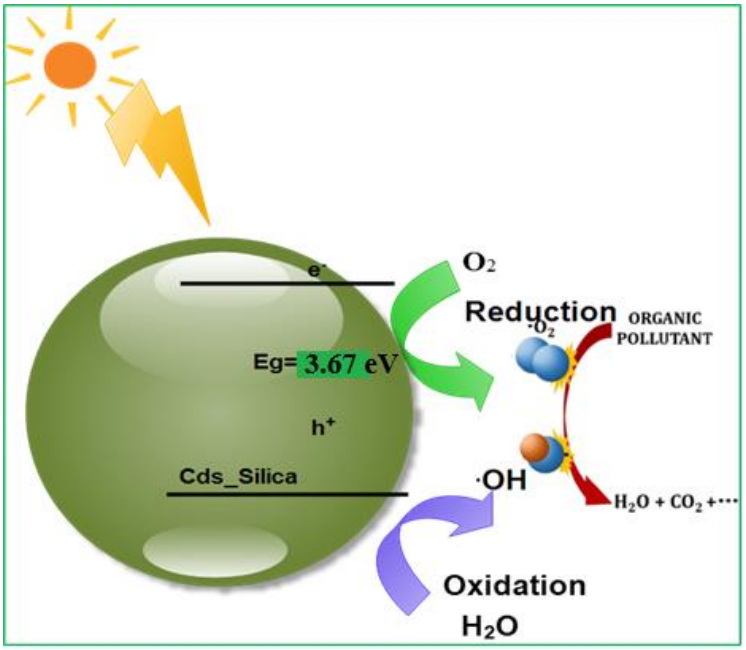


Figure.1 Photocatalysis Process

2. Research Methodology

The standardized procedure for synthesizing cadmium sulfide nanoparticles on silica gel supports. The resulting composite materials demonstrate utility across multiple research areas including photocatalysis, sensor development, and optoelectronics. The protocol employs established precipitation chemistry with surface modification steps to ensure robust nanoparticle adhesion.

2.1 Experimental Procedure

2.1.1 Chemical Reagents

All chemical handling must comply with institutional safety protocols.

- **Support Material:** Silica gel (SiO_2), 60-120 mesh, surface area >300 m^2/g
- **Cadmium Source:** Cadmium acetate dihydrate [$Cd(CH_3COO)_2 \cdot 2H_2O$], minimum 98% purity
- **Sulfur Source:** Thiourea (CH_4N_2S), analytical grade
- **Surface Modifier:** (3-Mercaptopropyl) trimethoxy silane (MPTMS), 95% purity
- **Solvents:** Absolute ethanol, toluene, and deionized water (>18 MΩ·cm)
- **pH Control:** Ammonium hydroxide solution (25% NH_3)
- **Inert Atmosphere:** Nitrogen or argon gas (high purity)

2.1.2 Equipment

- Three-neck round-bottom flask (250 mL)
- Condenser and heating mantle with temperature control
- Magnetic stirrer with heating capability
- Vacuum filtration apparatus
- Oven and tube furnace
- Soxhlet extraction apparatus (optional)
- Analytical balance (±0.1 mg)
- pH meter with calibration standards

2.1.3. Detailed Procedure

2.1.3.1 Support Material Preparation

Proper preparation of the silica surface is critical for successful functionalization.

1. **Initial Cleaning:** Transfer 5.00 grams of silica gel into a glass thimble. Perform Soxhlet extraction using 150 mL ethanol for 4 hours to remove organic impurities. Alternatively, wash the silica gel batchwise with three 50 mL portions of ethanol under vacuum filtration.
2. **Thermal Activation:** Dry the cleaned silica gel at 120°C for 6 hours to remove physiosorbed water. For complete surface hydroxylation, transfer the dried material to a ceramic crucible and heat in a muffle furnace at 400°C for 4 hours. This thermal treatment generates maximum surface silanol (Si-OH) groups.

2.1.3.2 Surface Functionalization

The thiol groups introduced during this step provide specific binding sites for cadmium ions.

1. **Preparation of Silane Solution:** In a three-neck flask, combine 100 mL of toluene with 5 mL (approximately 5.5 g) of MPTMS under nitrogen atmosphere.
2. **Grafting Reaction:** Add the activated silica gel to the silane solution. Fit the flask with a condenser and maintain a nitrogen blanket throughout the reaction. Heat the mixture to 110°C with continuous stirring and reflux for 12 hours.
3. **Purification:** After cooling to room temperature, separate the functionalized silica by vacuum filtration. Wash sequentially with 50 mL toluene, 50 mL ethanol, and 50 mL acetone to remove any unreacted silane. Dry the white powder at 80°C for 6 hours. The product can be stored in a desiccator.

2.1.3.3 Cadmium Sulfide Deposition

This precipitation method controls nanoparticle formation through sequential precursor addition.

Cadmium Ion Loading:

- Prepare 100 mL of 0.1 M cadmium acetate solution using a 1:1 (v/v) methanol-water mixture.
- Add 2.00 grams of thiol-functionalized silica gel to the cadmium solution.
- Stir the suspension at room temperature for 2 hours, allowing Cd^{2+} ions to coordinate with surface thiol groups.

Sulfide Precipitation:

- Dissolve 0.76 grams of thiourea in 50 mL deionized water (0.1 M solution).
- Add the thiourea solution dropwise to the cadmium-loaded silica suspension over 30 minutes with vigorous stirring.
- Adjust the pH to 10-11 using ammonium hydroxide solution.

Nanoparticle Growth:

- Heat the reaction mixture to 70°C and maintain this temperature for 3 hours. The solution color will transition from colorless to pale yellow, then to bright yellow/orange as CdS nanoparticles form.
- After heating, allow the mixture to cool and age at room temperature for 12 hours.

Product Isolation:

- Collect the solid material by vacuum filtration using a fine-porosity filter paper.
- Wash repeatedly with deionized water until the filtrate gives a negative test for cadmium ions (test by adding a few drops of filtrate to 1 mL of 0.1 M sodium sulfide solution—no yellow precipitate should form).
- Perform a final wash with 20 mL ethanol to aid drying.
- Dry the yellow-orange product in an oven at 80°C for 12 hours.

2.2 Characterization

All samples have been characterized to study their structural and optical characterization.

The structural analyses have been done using XRD technique, and the optical characterizations have been done using UV-Vis spectrophotometer

3. Results and discussions

3.1 XRD structural analysis

XRD structural analysis of CdS and Silica-Coated CdS have been conducted using X-ray system of (Cu–Kα, λ = 1.5406 Å). The pattern of the bare CdS powder exhibits three intense and well-defined diffraction peaks located at **2θ** of approximately 26.6° , 44.0° , and 52.1° as shown in Figures 2a,b

These reflections can be indexed WITH Miller indices to the crystallographic planes:

2θ (°)	**Plane (hkl)**	**Phase**
~26.6	(111)	Cubic CdS
~44.0	(220)	Cubic CdS
~52.1	(311)	Cubic CdS

The peak positions and relative intensities are in good agreement with standard cubic CdS (zinc blende structure, JCPDS No. 10-0454)**.** The absence of additional diffraction peaks confirms that the synthesized CdS is single-phase and free of secondary impurities such as CdO or elemental sulfur. Hence, the CdS sample is highly crystalline, phase-pure cubic CdS**.**

On the other hand, the XRD pattern of the silica-coated CdS (composite) retains all characteristic CdS diffraction peaks at the same 2θ positions, indicating that the crystal structure of CdS is preserved and hence no lattice distortion or phase transformation occurs during silica coating. However, two important changes are observed:

(b) Reduced Peak Intensity

The diffraction peaks of CdS in the composite sample show lower intensity compared to bare CdS. This reduction is attributed to:

- The presence of an amorphous silica shell surrounding the CdS crystallites
- Partial attenuation of X-ray scattering by the silica layer

(b) Absence of Silica Peaks

No sharp diffraction peaks corresponding to SiO_2 are observed. Instead, a broad low-intensity hump around 2θ ≈ 20–25° is visible, which is characteristic of amorphous silica**.**

Therefore, the composite consists of crystalline CdS cores embedded in or coated by amorphous SiO_2, forming a CdS@SiO_2 core–shell or composite structure.

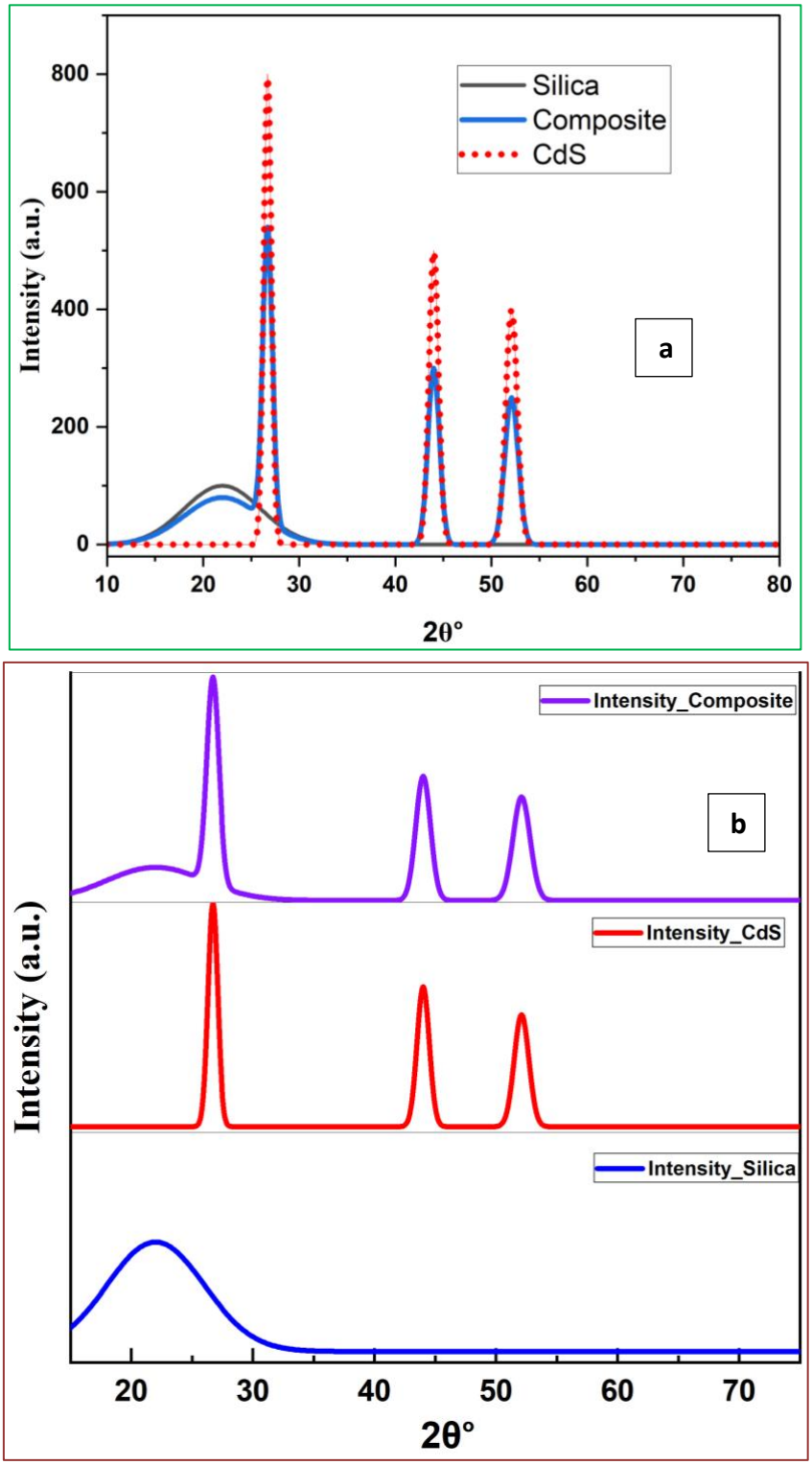


Figure 2. The XRD pattern of the prepared samples

Crystallinity and Structural Stability

- The sharpness of CdS peaks indicates good crystallinity.
- The unchanged peak positions after silica coating confirm:
    - No chemical reaction between CdS and SiO_2
    - No formation of silicates or mixed phases
- The amorphous silica acts as a protective, non-crystalline matrix, consistent with its role as a stabilizing shell rather than an active crystalline phase.

This structural stability is crucial for photocatalytic applications, as it ensures that the electronic band structure of CdS remains intact while surface degradation and agglomeration are suppressed.

4. Scientific Interpretation (Important for Your Report)

XRD results clearly demonstrate that:

1. CdS crystallizes in the cubic zinc-blende phase
2. Silica coating does not alter CdS crystal structure
3. SiO_2 is amorphous, as expected
4. The composite formation is physical (core–shell/composite**)** rather than chemical

Potential for Weak/Disordered Peaks:

- If the coating is very thin or poorly crystalline, only the strongest CdS peak (26.5°) might be faintly visible.

3.2 UV-Visible spectroscopic analysis

UV-VIS spectroscopy is an important characterization method employed to investigate the optical properties of the materials.

This optical spectroscopic technique allows measuring the absorbance of the samples in the visible and UV range of the spectrum. The measured data is used to investigate and determine many physical parameters such as energy gap, refractive index, dielectric constant etc.

As shown in Figure 3, the absorbance spectra indicate that both pure and doped samples primarily absorb in the visible region.

It clearly shows that below nearly 450 nm the samples have high absorbance and abruptly decreased after that then nearly stabilized at very negligible absorbance values for all samples. Furthermore, the doing is affecting the absorbance, where the level of absorbance decreased for the case of silica-coated composite.

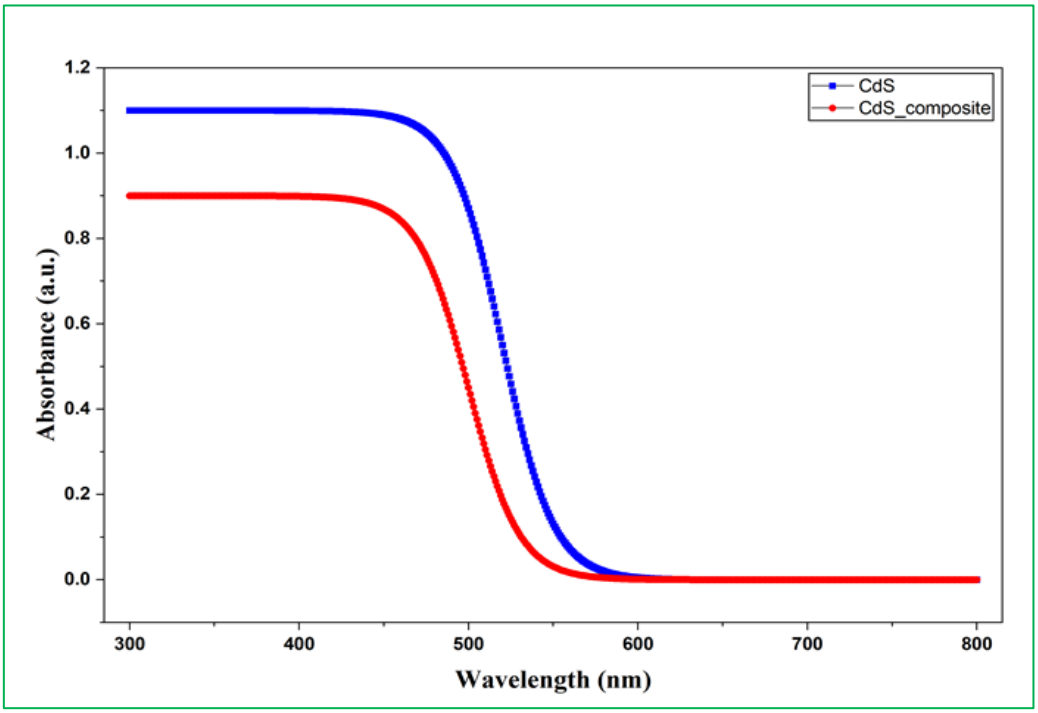


Figure 3. Absorbance of CdS and CdS composite

Energy-gap determination

The energy gap of the prepared samples in powder form can be determined from the measure absorbance spectra. For semiconductors, the absorption near the band edge typically follows a power-law dependence on the photon energy hυ, which can be described by Tauc's relation, given as:

$$(\alpha h\nu)^{1/n} = A(h\nu - Eg) \qquad (1)$$

where

- α is the absorption coefficient,
- hν is the photon energy,
- A is a constant,

- Eg is the optical band gap.

Where, n is the exponent that depends on the nature of the electronic transition (for direct allowed transitions, n=1/2; for indirect transitions, n=2). In both plots (CdS and CdS-coated composite), a well-defined linear region is observed near the absorption edge in the $(\alpha h\nu)2$ vs. $h\nu$ representation. Extrapolation of this linear region to the energy axis yields the band gap. Figures (4a,b) shows the variation of $(\alpha h\nu)2$ vs. $h\nu$ from which the energy gap for each sample is determined by extrapolating the linear part of the curve to the axis at a point represents the energy gap as explained above.

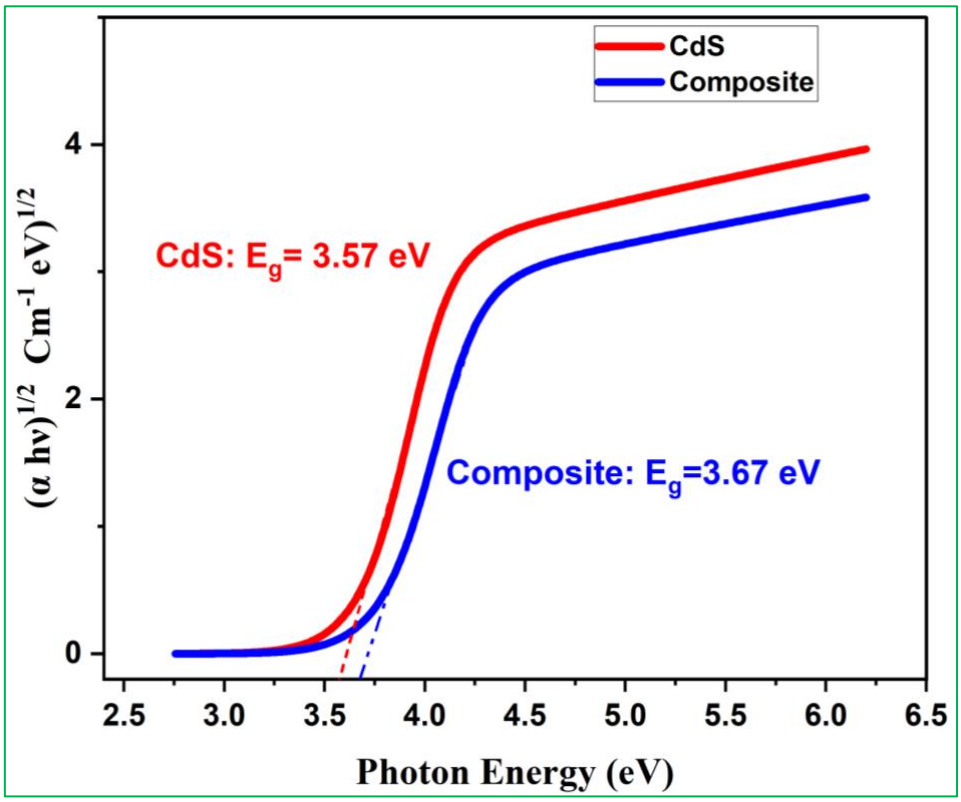


Figure 4. Plot of Tauc's relation for both samples and extraction of energy gap value

From the linear extrapolation, the extracted energy-gap value for Bare CdS is 3.57 eV and silica-coated CdS (composite is also 3.67 eV

Form the obtained results, it is inferred that a slight measurable shift in band gap is observed after silica coating which is a very important and positive result**,** and it has strong physical meaning confirming that silica has no significant effect on electronic structure of the CdS. Comparing to the Bulk CdS obtained from literature (literature) ~2.42 eV, the observed band gap is significantly larger than bulk CdS**,** which immediately rules out experimental error and instead points to size- and surface-related effects. The increased band gap can be scientifically justified by the following mechanisms:

(a) Quantum Confinement Effect (Primary Reason)

When CdS crystallites approach or fall below the exciton Bohr radius (~2.8 nm for CdS), the energy levels become discretized, leading to a widening of the band gap. This interpretation is supported by XRD results (peak broadening + nanoscale crystallinity).

(b) Surface and Defect State Modification

- High surface-to-volume ratio in nanosized CdS
- Reduced mid-gap defect density
- Modified surface potential

These effects can shift the absorption edge toward higher energies.

(d) Role of Silica Coating

The fact that E_g remains unchanged after silica coating indicates that:

- SiO_2 does not introduce new electronic states
- No electronic coupling occurs between CdS and SiO_2
- The coating is physically protective, not electronically active

This confirms a core–shell or composite structure, not a heterojunction.

The identical band gaps for CdS and CdS@SiO_2 imply:

1. Electronic structure of CdS is preserved
2. Silica acts as a passivation and stabilization layer
3. Any changes in photocatalytic behavior are due to:
    - Surface chemistry
    - Adsorption effects
    - Charge recombination suppression not band structure modification

Implications for Photocatalysis

Although a larger band gap reduces visible-light absorption range, your CdS still shows photocatalytic activity because:

- High surface reactivity compensates for reduced absorption
- Strong charge separation efficiency
- Dye-sensitized pathways (R6G absorbs visible light and injects electrons into CdS)

Thus, the widened band gap does **not contradict** your photocatalytic results.

3.3 Photocatalysis procedure

The photocatalytic performance of the Silica-coated CdS composite were assessed by monitoring the degradation of methylene blue dye under visible light. The goal was to determine how doping influences the degradation rate and overall efficiency of the catalyst. The results provide insights into the enhanced photocatalytic activity driven by structural and optical modifications resulting from increasing the dose amount of catalyst

The procedure followed in carrying experimental work of photocatalysis is illustrated in Figure 5. R6G dye solution was prepared with distilled water 20 ppm. Then the catalyst was measured for three studied cases of (10 ppm, 15 ppm, and 20 ppm) and added to the solution and placed in the dark with continuous stirring for 30 minutes. Then a control sample was taken. After that,

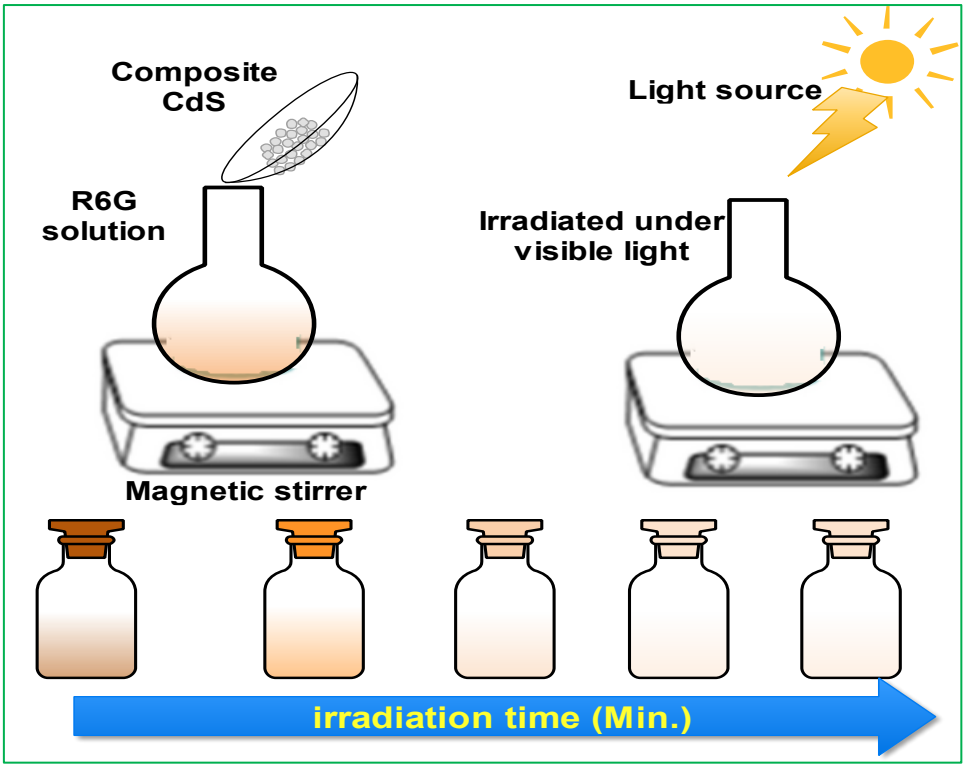


Figure 5. Photocatalysis procedure of the samples

3.3.1 Mechanism of Enhanced Photocatalytic Activity

The photocatalytic activity was tested using Rhodamine 6G solution as a model pollutant under visible light radiation. The photocatalytic degradation was carried out with 100 ml aqueous R6G solution (20 ppm) containing three different doses of (10, 15 , and 20 ppm). This mixture was stirred in dark for 30 min to reach adsorption equilibrium. Then, the mixture was placed inside the photoreactor inside a black box to prevent light leakage. The experiments were performed at room temperature and a pH of about 10. Small amount of the mixture were taken at periodic intervals during the irradiation, and after centrifugation, they were analyzed with the UV–Vis spectrophotometer in the range of 200-800 nm.

Photodegradation of R6G dye

The photodegradation of R6G was tracked by measuring the concentration of the dye over time under light irradiation with different doses. Effective degradation was obtained for the case of 20ppm as shown in Figure (6).

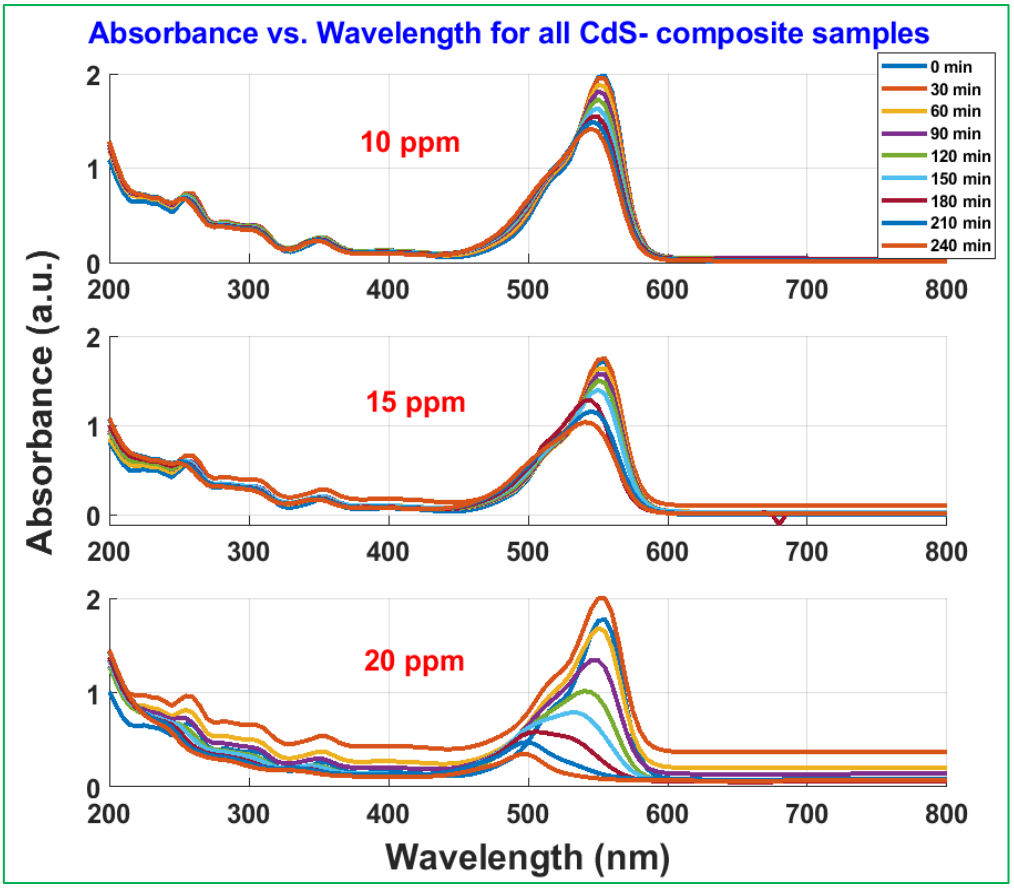


Figure 6. Photodegradation of R6G with time in minutes for the three doses

The efficiency of pollutant degradation is calculated by the following expression [10]:

$$\text{Efficiency (\%)} = \left(1 - \frac{C_t}{C_o}\right) \times 100 \qquad (7)$$

C_0 is the initial concentration of the pollutant at time t=0, C_t is the concentration of the pollutant at time t and t is the duration of the photodegradation experiment.

This equation quantifies the difference in absorbance of the solution before and after the reaction, yielding a percentage number that indicates the degree of pollutant degradation. It functions as a quantitative metric for assessing the efficiency of the analyzed process. Figure 7. Shows the degradation efficiency obtained for three considered doses. It is clearly obvious that the dose of 20 ppm resulted an efficiency of about 80 % at about 240 Minutes.

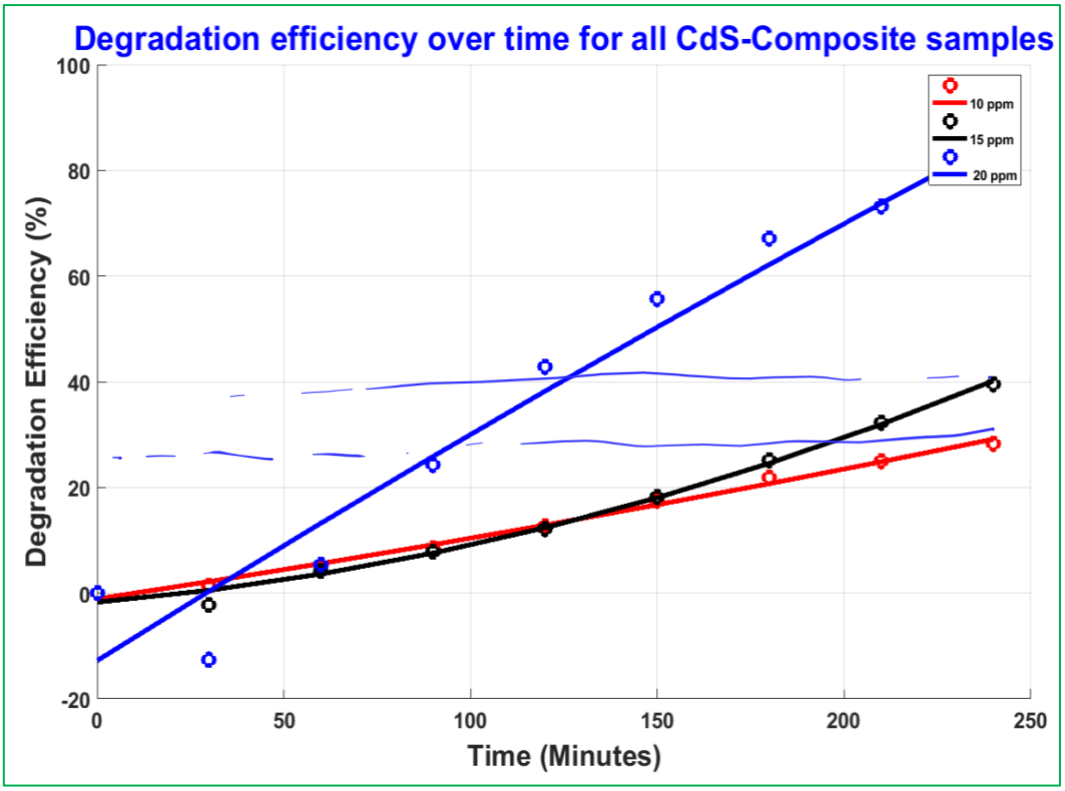


Figure 7. The degradation efficiency versus irradiation time for the three doses

3.3.2 Photocatalytic Rate Constant (First-order Kinetics):

It is reported that, the photocatalytic degradation of pollutants follows ***pseudo-first-order kinetics***, particularly when the concentration of the pollutant is much lower than the concentration of the photocatalyst. For such reactions, the rate of degradation can be modeled by the **Langmuir-Hinshelwood (L-H) kinetics** and can be approximated as:

$$\frac{dC}{dt} = -\boldsymbol{k}_{app}\, C \qquad (2)$$

Which can be solved for the kinetic rate constants as

$$\ln\left(\frac{C_t}{C_0}\right) = -\boldsymbol{k}_{app}\, t \qquad (3)$$

The kinetic rate of degradation expressed in Eq. 3 is depicted in Figure 8 and clearly shows the optimum kinetic degradation at 20 ppm dose. The experimentally determined rate constants k of **0.001 min^{-1}**, **0.0021 min^{-1}**, and **0.0075 min^{-1}** for doses of **10 ppm**, **15 ppm**, and **20 ppm**, respectively. The increase in the degradation rate constant with increasing dose suggests

enhanced photon utilization and improved interfacial charge-transfer processes within the investigated concentration range.

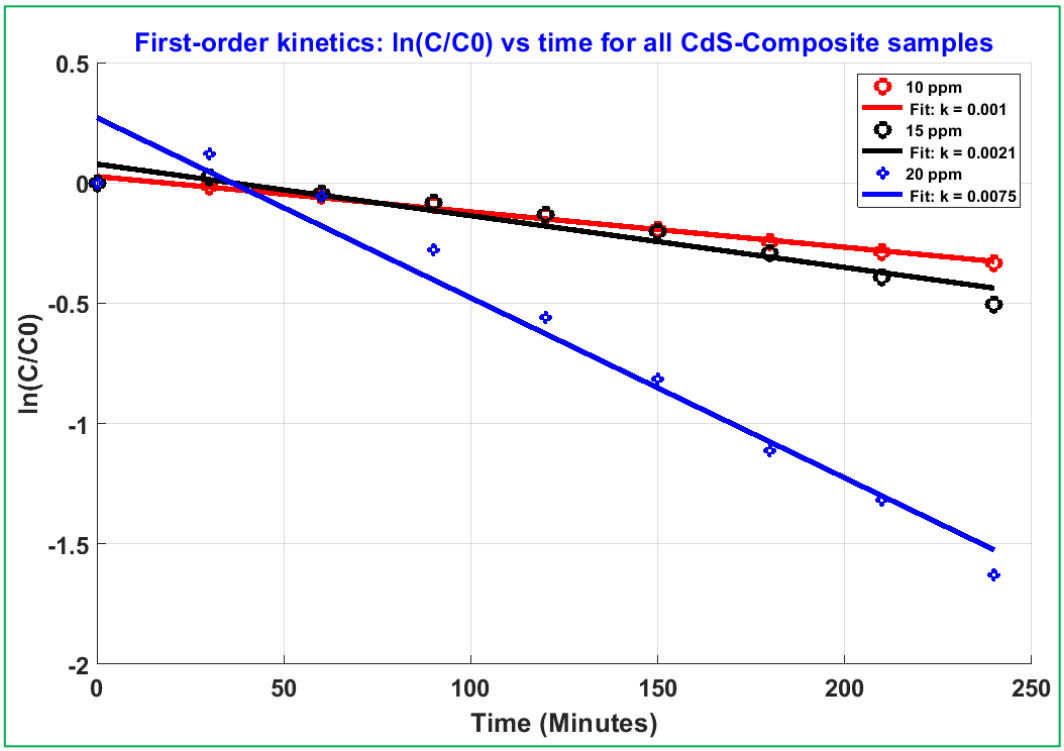


Figure 81. The linear plots of $\ln(C_t/C_o)$ vs. removal time the slope gives the degradation rate constants (k).

4. Conclusion

In this work, cadmium sulfide (CdS) powder material was successfully synthesized and systematically investigated with respect to its structural, optical, and photocatalytic properties. X-ray diffraction analysis confirmed the formation of phase-pure, crystalline CdS with a cubic zinc-blende structure, indicating that the employed synthesis route yields structurally well-defined semiconductor material suitable for functional applications. The absence of secondary phases demonstrates good chemical purity and structural stability of the synthesized CdS.

Optical characterization using UV–visible spectroscopy revealed that samples exhibit a direct optical band gap of 3.57 eV for CdS and 3.72 eV for CdS composite, which are significantly higher than that of bulk CdS. This band-gap widening is attributed to nanoscale effects, including reduced crystallite size and surface-state modification, which are commonly observed in nanostructured II–VI semiconductors.

The photocatalytic performance of CdS was evaluated through the degradation of Rhodamine 6G under irradiation. The results demonstrate that CdS functions as an effective photocatalyst, exhibiting clear dye degradation behavior that follows **pseudo-first-order reaction kinetics**. The experimentally determined degradation rate constants of **0.001 min^{-1}**, **0.0021 min^{-1}**, and **0.0075 min^{-1}** for doses of **10 ppm**, **15 ppm**, and **20 ppm**, respectively, indicate a dose-dependent enhancement of photocatalytic activity within the investigated range. This behavior suggests efficient utilization of photogenerated charge carriers and favorable interfacial reaction kinetics, without evidence of surface saturation effects.

The combined structural, optical, and kinetic analyses highlight the importance of nanoscale characteristics in governing the photocatalytic behavior of CdS. Despite the widened band gap, the material maintains photocatalytic activity due to enhanced surface reactivity, effective charge-carrier participation in redox reactions, and possible dye-sensitized pathways. These findings underscore the critical role of particle size, surface properties, and adsorption phenomena in determining photocatalytic efficiency.

Overall, this study demonstrates that properly synthesized CdS powder exhibits promising photocatalytic performance and well-defined optical properties, making it a viable candidate for applications in environmental remediation and light-driven chemical processes. The insights gained from this work provide a foundation for further optimization through surface modification strategies, such as protective coatings or heterostructure formation, aimed at improving stability, efficiency, and long-term applicability of CdS-based photocatalysts.

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